

CHENSHUHAO (CODY) QIN

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EDUCATION

- University of Southern California (USC)**
Master of Science in Computer Science
Los Angeles, CA
Jan. 2026 – Dec. 2027 (Expected)
- Coursework:** Machine Learning, Deep Learning, Natural Language Processing, Data Structures & Algorithms, Operating Systems, Computer Architecture, Computer Networks, Predictive Modeling
- Duke University; Duke Kunshan University**
Bachelor of Science in Computer Science; B.S. in Computation and Design
Durham, NC / Kunshan, China
Aug. 2021 – May. 2025
- Honors:** Apple WWDC23 Swift Student Challenge Winner (1 of 13 in China), Provincial Innovation Grant Recipient (\$3k), DKU Student Ambassador

SKILLS

- Languages:** Python, TypeScript/JavaScript, Java, SQL, Swift, C
- AI & Machine Learning:** PyTorch, TensorFlow, LoRA Fine-tuning, Prompt Engineering, Hugging Face, Pandas, NumPy
- LLM & Agents:** LLM Agents (AgentScope, LangChain, LlamaIndex), Agent Harness & Context Engineering, Retrieval-Augmented Generation (RAG), Model Serving (vLLM), Vector Search (Milvus)
- Infrastructure & Distributed Systems:** Ray, Kubernetes, Docker, MySQL, PostgreSQL, Redis, REST APIs (FastAPI, Flask), React, Linux, Git, CI/CD, Object Storage (OSS/S3), AWS, PyTest

WORK EXPERIENCE

- Alibaba Cloud (Top-4 Global Cloud Provider)**
AI Engineer Intern
Hangzhou, China
Jun. 2026 – Aug. 2026
- Collaborated with Alibaba’s **Tongyi Lab** to productize **Data-Juicer Agents** — an open-source **agent harness** — into natural-language-driven multimodal data pipelines (cleaning, filtering, deduplication) for the broader **Qwen** ecosystem, replacing script-heavy workflows on the Apsara Data Lake platform.
 - Built an agent-to-Ray distributed execution pipeline over private-cloud object storage (OSS) and Kubernetes — **2,793 records, 16 operators, ~5 minutes** — and landed the core capability upstream through a merged PR.
 - Designed a three-layer context-control architecture (tool-output compaction, memory compression, hard token budgets) cutting single-tool context payloads by **82%** and enabling stable multi-turn agents on 30k-token models.
 - Shipped a full-stack agent workspace (**React / FastAPI / MySQL**, REST APIs) with persistent sessions, execution tracing, conversation rollback, and multi-instance isolation; delivered **4 production container-image releases**.
 - Contributed **5 merged PRs** (PyTest unit coverage, CI green) to the 7k+ Data-Juicer open-source ecosystem across multimodal ingestion, distributed Ray execution, and context optimization; 1 enhancement proposal adopted by maintainers.
- Blueberry AI**
AI Engineer Intern
Burlingame, CA (Remote)
Nov. 2025 – Jan. 2026
- Built a **Milvus + MySQL** feature-extraction and auto-tagging pipeline batch-indexing **200K+ images** for landmark / IP / SKU retrieval; benchmarked **DINOv3**, **Qwen-VL**, and **BGE-VL** under a unified recall / precision / latency framework for private deployment; delivered a **YOLO + Milvus** vector search + Gradio retrieval POC validated for commercial use.
- Tsinghua Sichuan Energy Internet Research Institute**
Algorithm Engineer Intern
Chengdu, China
Sep. 2025 – Oct. 2025
- Engineered a **DeepLabV3+ (PyTorch)** semantic-segmentation model for microgrid fault diagnosis reaching **92% accuracy** across complex field scenarios; owned the full data lifecycle of **500+ annotated images** from preprocessing to deployment, anchoring winning bids for major energy-infrastructure projects.
- Leiting Games (G-bits)**
AI R&D Intern (AIGC & Automation)
Shenzhen, China
May. 2025 – Aug. 2025
- Fine-tuned **SDXL / Flux LoRA** models on **Kohya**, shipping 30+ ad assets (80% adoption) that drove **117K+ installs** and **¥24.6M billings (196% ROI)**; led a 5-person team scaling creative output **3×** with 15–20% higher material ROI; built a **98.85%-accurate** RPA monitor catching 1,000+ private-server violations (2025 Best Practice).

PROJECTS & PUBLICATIONS

- ChatDKU — RAG Campus Q&A Assistant** | *Python, LlamaIndex, LangChain, Redis*
Jul. 2024 – May. 2025
- Co-founded a university-adopted **Retrieval-Augmented Generation (RAG)** Q&A agent indexing **12,000+ webpages** over a privately deployed LLM; implemented **Hybrid Search (BM25 + vectors)**, **HyDE**, multi-level **Redis** caching, and **SSO** web deployment; the system was officially deployed at DKU and recognized by multiple university offices, documented as a DKU Signature Work.
- Publication:** *Privacy-Preserving Room Occupancy Estimation Using Federated Analytics of BLE Packets* — **ACM SenSys 2024** (Demo Track, second author), pp. 889–890; **95%** real-world accuracy via federated learning over BLE.